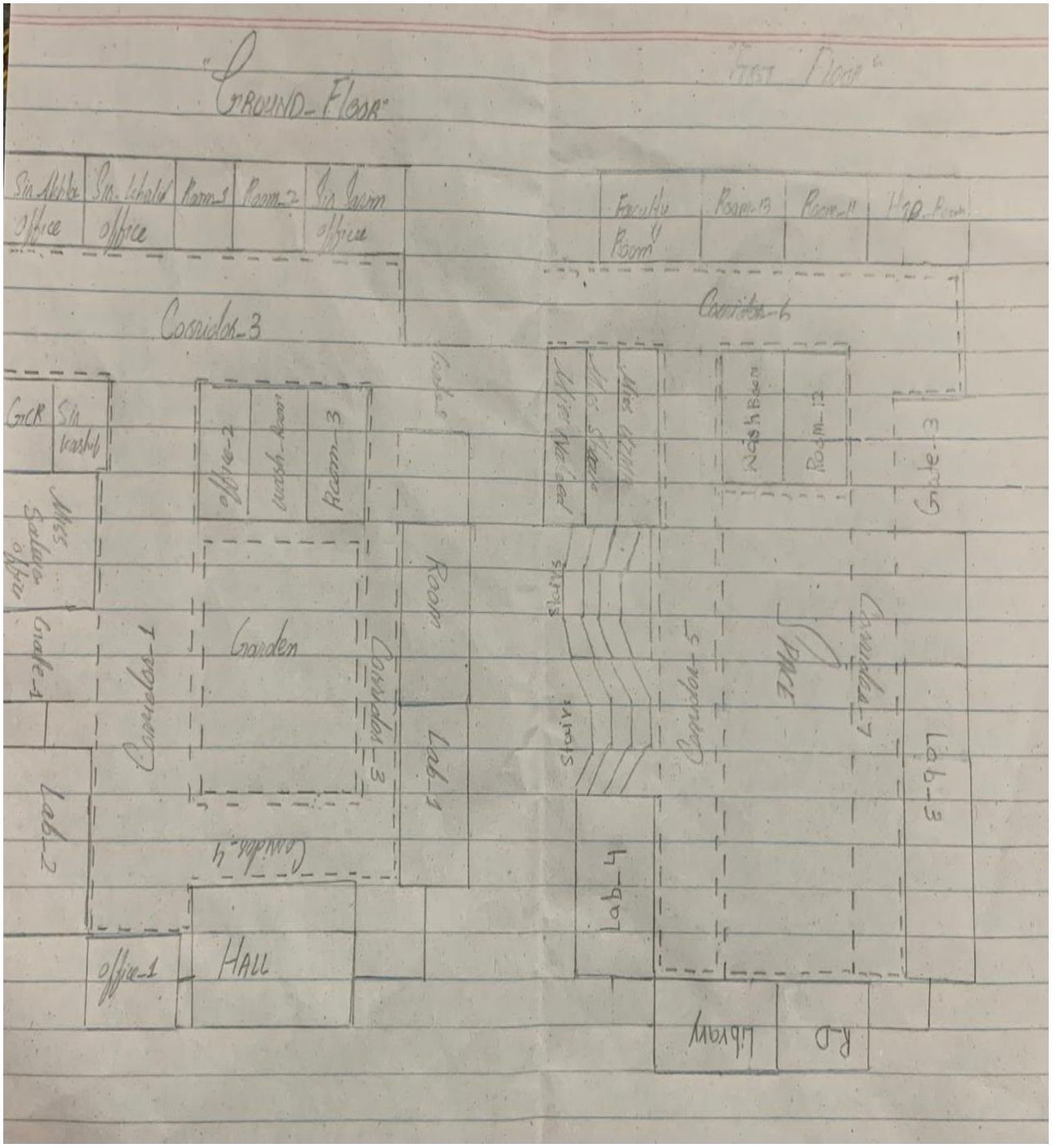

"GROUND-Floor"
"FIRST Floor"
Sir Akbar Office
Sir Khalid office
Room-1
Room-2
Sir Jaam office
Cosridor-3
GICR
Sir Wasib
Office-2
wash Room
Room-3
Mens Gallery after
Garden
Corridor-1
Grade-1
Lab-2
Corridor-2
office-1
HALL
Corridor-3
Room
Lab-1
Faculty Room
Room-13
Room-14
Hod Room
Corridor-6
Mens Gallery
Mens Staff
Wash Room
Room-12
Grade-3
Stairs
Corridor-5
Stairs
Lab-4
Corridor-7
Lab-3
Library
RO

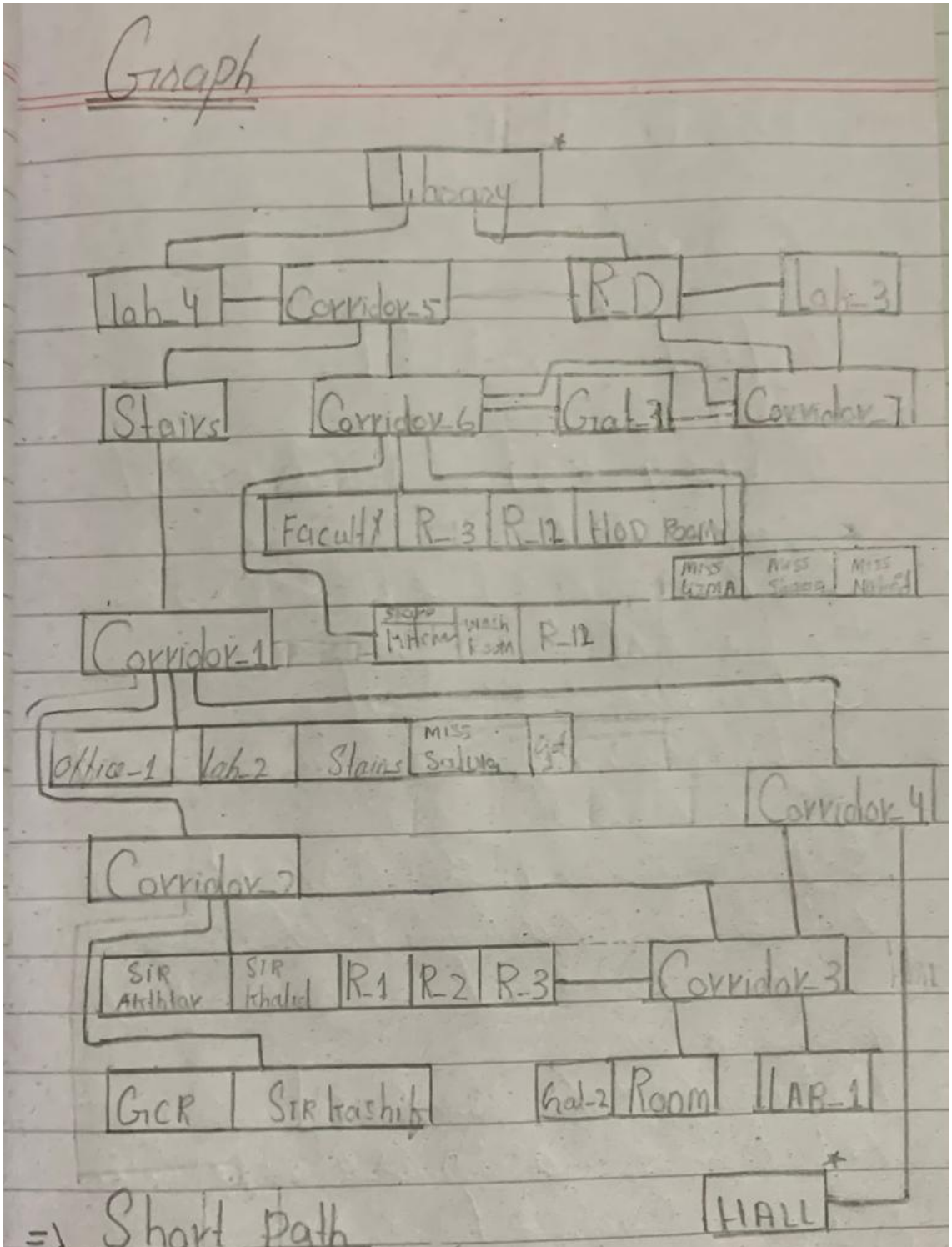
# Ground Floor

Lab-1 → Corridor-3
Lab-2 → Corridor-1
Room-1 → Corridor-2
Room-2 → Corridor-2
Room-3 → Corridor-1
Sir Sawm R → Corridor-2
Sir Akhlan R → Corridor-2
Sir Khalid R → Corridor-2
Office-1 → Corridor-1
Office-2 → Corridor-2
GCR → Corridor-2
Miss Salwa R → Corridor-1
Gal-1 → Corridor-1
Gal-2 → Corridor-3
Hall → Corridor-4, Corridor-1
Stair → Corridor-5, Corridor-1
Corridor-1 → Office-1, Lab-1, Miss Salwa, Stair, Ga
Corridor-2 → Room-1, Room-2, Sir Sawm, GA, S.K, G2, S.h
Corridor-3 → Lab-1, Gal-2, Corridor-2, Corridor-4,
Corridor-4 → Hall, Corridor-1
Sir Kashif → Corridor-2

# FIRST FLOOR

Library → Lab-4, R.D
Lab-4 → R.D, Corridor-5
Lab-3 → Corridor-7, R.D
HOD-R → Corridor-6
Faculty-R → Corridor-6
Miss Hamna-R → Corridor-6
Miss Shania-R → Corridor-6
Miss Nabeed-R → Corridor-6
Girls Washroom → Corridor-6
Room-11 → Corridor-6
Room-12 → Corridor-6
Room-13 → Corridor-6
Conf-3 → Corridor-6
R.D → Corridor-5
Corridor-5 → lab-4, stairs, Corridor-6, R.D
Corridor-6 → R.D, HOD.R, Faculty-R, Miss Hamna-R, Miss-Nabeed-R, Miss Shania-R, Girls-W.R, Room-11 Room-12, Room-13, Conf-3,
Corridor-7 → lab-3, R.D, Corridor-6,

# Graph




## => Short path

*Library → Lab_4 → Corridor_5 → Stairs → Corridor_1 → Corridor_4 → HALL.

Lib to Hall

Inqueue
Lib
Q: [Lib] , V: [Lib]
Dequeue Lib
Q: [Lab-4, R-D] , V: [Lib, lab-4, R-D]
Dequeue lab-4
Q: [R-D, Corridor-5] , V: [Lib, lab-4, R-D, Corridor
Dequeue: R-D
Q = [corridor5, lab-3, Corridor-7]
V = [Lib, Lab-4, R-D, Corridor5, lab-3, Corridor-7]
Dequeue: Corrido-5.
Q = [lab-3, Corridor-7, Stairs, Corridor-6]
V = [Lib, lab-4, R-D, Corridor-5, Lab-3, Corridor-7, Stairs, Corridor-6]
Dequeue lab-3
Q = [Corridor-7, Stairs, Corridor-6]
V = [Lib, lab-4, R-D, Corridor-5, lab-3, Corrido Stairs, Corridor-6]
Dequeue: Corridor-7
Q = [stairs, Corridor-6]
V = [Lib, lab-4, R-D, Corridor-5, lab-3

Corridor-7, Stairs, Corridor-6 ]

- Destination: Stairs
  O: [ Corridor-6, Corridor-7 ]
  V: [ Lib, lab-4, R-D, Corridor-5, lab-3, Corridor-7, Stairs, Corridor-6, Corridor-1 ]
- Destination: Corridor-6
  O: [ Corridor-1, Faculty, R-13, R-12, HOD, Cab-3, Miss Uzma, Miss Sharia, Miss Nabeel, Girls washroom ]
  V: [ Lib, lab-4, R-D, Corridor-5, lab-3, Corridor-7, Stairs, Corridor-6, Corridor-1, Faculty, R-13, R-12, R-11, Cab-3, Miss Uzma, Miss Sharia, Miss Nabeel, Girls washroom ]

Destination: Corridor-1
O: [ Faculty, R-13, R-12, HOD, Cab-3, Miss Uzma, Miss Sharia, Miss Nabeel, Girls washroom, office-1, lab-2, Stairs, Miss Salwa, Corridor-2, Corridor-4 ]
V: [ Lib, lab-4, R-D, Corridor-5, lab-3, Corridor-7, Stairs, Corridor-6, Corridor-1, Faculty, R-13, R-12, R-11, Cab-3, Miss Uzma, Miss Sharia, Miss Nabeel, Girls washroom, office-1, lab-2, Stairs, Miss Salwa, Corridor-2, Corridor-4 ]

Sequence: Faculty, R-13, R-12, R-11, Gal-3, HOD, Miss Uzma, Miss-Shazia, Miss-Nabeel, washroom.

O = [ office-1, lab-2, Stair, Miss Saliha, Corridor-2, Corridor-4 ]

V = [ Lib, lab-4, R-D, Corridor-5, lab-3, Corridor-7, Stair, Corridor-6, Corridor-1, Faculty, R-13, R-12, R-11, Gate-3, Miss Uzma, Miss Shazia, Miss-Nabeel, Girls washroom, office-1, lab-2, Stair, Miss Saliha, Corridor-2, Corridor-4 ]

1. Sequence = office-1, lab-2, Stair, Miss-Saliha

O = [ Lib, lab-4, R-D, Corridor-5, lab-3, Corridor-7

V = [ Stair, Corridor-6, Corridor-1, Faculty - R-13, R-12, R-11, Gate-3, Miss-Uzma, Miss-Shazia, Miss-Nabeel, Girls-washroom, office-1, lab-2, Stair, Miss-Saliha, Corridor-2, Corridor-4 ]

1. Sequence: Corridor-2

O: [ Corridor-4, Sir-Akhtar, Sir-Khalil, R-1, R-2, R-3, G-CR, Sir-Yasir, Corridor-3 ]

V = [ Lib, lab-4, R-D, Corridor-5, lab-3, Corridor-7, Stair, Corridor-6, Corridor-1, Faculty, R-13, R-12, R-11, Gate-3, Miss-Uzma, Miss-Shazia, Miss-Nabeel, Girls-washroom, office-1, lab-2

Stairs, Miss Saba, Corridor-2, Corridor-4, Sir Akh, Sir-Kashif, R-1, R-2, R-3, CoCR, Sir-Kashif Corridor-3]

. Derivation: Corridor-4

O = [Sir-Akhtar, Sir-Khalid, R-1, R-2, R-3, CoCR, Sir-Kashif, Corridor-3, Hall,

V = [Lib, Lab-4, R.D, Corridor-5, Lab-3, Corridor-7, Stairs, R-11, Cal-3, Miss-Uzma, Miss Sharia, Marshal, Girls wash room, office-1, Lab-2, Stairs, Miss Saba, Corridor-2, Corridor-4, Sir-Akhtar, Sir Kashif, R-1, R-2, R-3, CoCR, Sir-Kashif, Corridor-3 Hall]

. Derivation: Sir-Akhtar, Sir Khalid, R-1, R-2, R-3, CoCR Sir-Kashif,

O = [Corridor-3, HALL]

V = [ , , , , , , , ]

. Derivation: Corridor-3

O = [HALL, Cal-3, Room, Lab-1]

V = [Lib, Lab-4, R.D, Corridor-5, Lab-3, Corridor-7, Stairs, R-11, Cal-3, Miss-Uzma, Miss Saba, Miss Sharia, Miss Nabeel, Girls wash room, office-1, Lab-2, Stairs, Miss Saba, Corridor-2, Corridor-4, Sir-Akhtar, Sir Kashif, R.D, R-2, R-3, CoCR, Sir Khalid, Cafe-1, Corridor-3, Hall, Cafe-2, Room, Lab-1]

1 Departure: Hall

O: [Gal-2, Room lab-1]

V: [Lib ..... ------ Hall --- Gal-2, Room lab-1]

1 Departure: Gate-2, Room, lab-1

O: [ ]

V: [Lib, lab-4, R-10, Corridor-5, lab-3, Corridor-1, Stairs, Corridor-6, Corridor-1, Faculty, R-13, R-12, R-11, Girl wash-room, office-1, Gal-3, Miss-Uzma, Miss-Sharoo, Miss Nabeel, lab-2, Stairs, Gal-1, Mir-Salwa Corridor-2, Corridor-4, Sir-Ikram, Sir-Khalid, Sir-Hasnain, R-1, R-2, R-3, GKR, Corridor-3, HALL, Gal-2, Room, lab-1]